ANDREW AMAVIZCA

Robotics & Systems Engineer | Autonomous Sensing & Perception

US Citizen | aamavizca94@gmail.com | 1(209)769-4216 | Merced, CA | Open to relocate | LinkedIn | Portfolio

EDUCATION

University of California, Merced — B.S. Mechanical Engineering, May 2025

Focus: Controls, Embedded Systems, Systems Integration, Data-Science

CORE SKILLS

Programming: Python, C++, Embedded C/C++, MATLAB

Autonomy & Robotics: perception pipelines, sensor integration, autonomous sensing systems, health monitoring, anomaly detection, V&V

Sensors & Hardware: Remote sensing, cameras, embedded sensors, IoT devices, unmanned systems (UGV/UAV), modular sensor platforms

Machine Learning & Data: Perception-focused ML, explainable AI (XAI), large-scale sensor data analysis

Systems: Systems integration, requirements-driven design, field testing, automation tooling, cross-functional collaboration

EXPERIENCE

ASSOCIATE MANUFACTURING ENGINEER — Applied Aerospace & Defense, Stockton, CA | Oct 2025 - Present

- Developed Python automation tools that eliminated repetitive engineering work, recovering ~5 engineer-hours/day per engineer and saving tens of thousands annually
- Interpreted engineering drawings, Conditions of Supply, and EO/NIEO data to define system interfaces, tolerances, and manufacturability constraints for complex aerospace assemblies
- Built complete BOMs and performed make-vs-buy analyses to support supplier quoting, cost decisions, schedule, and risk reduction
- Authored test procedures, acceptance criteria, and temperature-controlled material-handling instructions for qualification-sensitive components
- Worked cross-functionally with technicians, quality, and supply chain teams to resolve system-level hardware blockers and sustain production throughput under tight schedules

SELECTED PROJECTS

AUTONOMOUS FLOOD IRRIGATION MONITORING SYSTEM – End-to-End System Owner

- Owned end-to-end design, integration, and deployment of an autonomous, field-deployed sensing system operating unattended in real-world environments
- Integrated multiple sensors with embedded control logic to meet real-world reliability and response constraints
- Implemented system health monitoring including battery status, signal integrity, and fault detection
- Conducted real-world field testing and validation to ensure reliability and operational readiness

MODULAR UGV SENSOR PLATFORM – Systems Integration

- Designed modular sensor platforms for unmanned ground vehicles supporting scalable integration of heterogeneous sensors
- Collaborated with mechanical and software teams to validate integrated systems under operational conditions
- Defined standardized mechanical, electrical, and data interfaces to enable rapid sensor swaps and future expansion

PERCEPTION & REMOTE SENSING PIPELINES (UAV / SATELLITE) – Perception & Data Systems

- Developed perception pipelines using machine learning to process and analyze noisy real-world sensor data
- Applied explainable AI techniques to improve model interpretability, validation, and failure analysis
- Validated perception outputs through iterative testing with real-world data

ADDITIONAL

Eligible for U.S. Secret clearance | Comfortable with field testing and on-site integration.